

COURSE CURRICULUM

BASIC CONCEPTS

1. Introduction to Python (What makes Python Best of the Best ?)

2. Basic Operations in Python

3. Variables

4. Data Types

- None
- Numeric
- String
- List
- Tuple
- Set
- Dictionary
- Range

5. Operators in Python

6. Bitwise Operators (Background Mechanism)

7. Number Systems Conversions

8. Swapping Variables

9. Math functions in Python

10. User input in Python

11. Conditional Statements in Python

12. Loops

13. Loop Control Statements

INTERMEDIATE CONCEPTS

1. Arrays

2. Using Numpy in Arrays

3. Copying Array & Matrix in Python

4. Functions in Python

5. Function Arguments and Types

6. Keyworded variable Length Arguments(**kwargs)

7. Global Keyword

8. Fibonacci Series and Factorial.

9. Recursion

10. Lambda Function

11. Filter, map, reduce

12. Decorators

13. Modules

14. Special Variable (__name__)

ADVANCE CONCEPTS

1. OOPs concept in Python (class / object)

2. Constructor

3. Types of Variables and Methods

4. Inner class

5. Inheritance and Constructor in Inheritance

6. Polymorphism

- Duck Typing
- Method Overloading
- Method Overriding
- Operator Overloading

7. Abstraction

8. Encapsulation

9. Exception Handling

10. File Handling

11. Searchings and Sortings